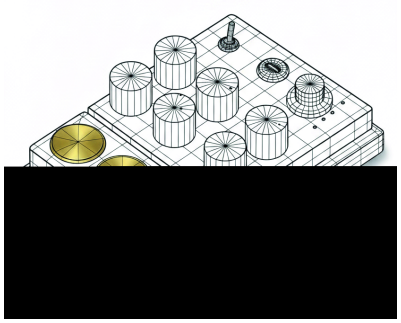
// generative instrument

Kac_OS Bass

diceFriend / quick start



Delta-sigma / Lorenz phrases and an eight-region FM/PM morph matrix. Audio is two octaves lower.

START: USB-C power + 3.5 mm stereo audio out. Connect to a mixer or speakers; begin with the listening level low.

AS: evolving delta-sigma phrases. **TS:** KACC / Metal / Fold / Grind.

AS: T1 freezes pitch; T2 widens the walk; T3 adds glitch / fold. **TS:** three voices; pads have priority over incoming MIDI.

// KNOBS / AS

P	NORMAL	SHIFT
1	Tempo / MIDI transpose*	Fine ratio / detune
2	Modulator ratio	Cross-feedback
3	FM / PM depth	Mutation probability
4	Wavefolder	Pattern length: 2-16
5	Harmonic stretch	Smoothing / pitch range
6	Morph position	Saturation

// KNOBS / TS

P	NORMAL	SHIFT
1	T1 pitch	FM / PM depth
2	T2 pitch	Phase grit / feedback
3	T3 pitch	Modulator ratio
4	T1 decay	Wavefolder
5	T2 decay	Phase flip
6	T3 decay	KACC morph

SHIFT: up = second layer; down = normal. **Pickup:** After changing layers or using DICE, move a knob to take control of its stored value.

// DICE / SHIFT

Tap DICE: randomize AS while preserving tempo, or select the next TS engine. **SHIFT + DICE 3 s:** randomize controls. **DICE 3 s:** AS / TS. **SHIFT + DICE 10 s:** bootloader.

// CV IN

CV1: trigger / clock. **CV2:** modulation; repeated pulses can become clock / triggers. In TS, external triggers allocate voices.
Banana: 0-9 V; common GND.

// USB MIDI

IN: notes with three-voice TS allocation and Note Off.
OUT: AS notes; TS pads on channels 1 / 2 / 3; pressure and CV2 pitch bend. Musical MIDI Thru.
Panel: CC20-25 = normal P1-P6; CC26-31 = SHIFT P1-P6.

*With external clock, normal P1 transposes the second MIDI-output layer. TS pitch knobs detune MIDI by about +/-1 octave. Bass audio is shifted down two octaves, with a 20 Hz floor.
